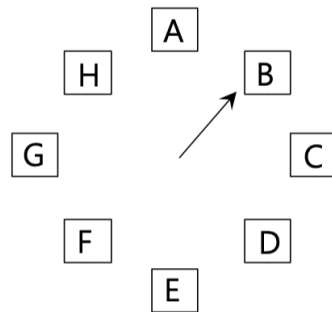


Name: _____

Assume main memory originally holds all pages in a clock, as shown in the image below:



All pages initially have clock bits flipped to 0. When a page is accessed, the bit is flipped to 1. The clock starts pointing to page B. Assume the following pages need to be read from (or written to) in the given order : D B C F B F A G

1. For each page, list the value held by its clock bit.
2. Assume we now need to load page I. We will do so by running the page replacement algorithm clock LRU. The clock hand rotates clockwise. If a page has a clock bit of 1, it flips the bit to 0 and then moves on. If it finds a page with a clock bit of 0, it removes that page. Which page is removed?
3. For each page, list the value held by its clock bit after the page replacement algorithm above removed a page (and replaced it with I). Assume page I has a clock bit of 1 as it has been accessed.
4. What are the benefits of running the clock LRU algorithm over standard LRU?